

TANVI AGARWAL

✉ tanvi.agarwal217@gmail.com ☎ +1 (413) 437-6272 🌐 github.com/tanvi217 🔗 linkedin.com/in/tanvi217

WORK EXPERIENCE

Tesla, Software Engineer Intern - Core Services

Aug 2025 – Aug 2026 | California, USA

Technologies: Go, Kafka, Redis, MySQL, Elasticsearch, Kubernetes, gRPC, Grafana

- Developed a Kafka-based Robotaxi payment dispute pipeline in Go, using concurrent consumers, Redis-based distributed locks, idempotency checks and DLQ handling to process 10K+ payment events/day.
- Built a job-based fare-data aggregator for pricing model training; ingests ride-quote Kafka events for stratified temporal & spatial sampling across H3 hexes; Redis token-bucket rate limiting (50K/month), calls external APIs.
- Created the e2e MySQL to ES CDC workflow using Debezium; binlog capture, connector configuration and indexing; replicating 100+ row changes/s; cut ride payment history p95 search latency from 2+ minutes to under a second.

Microsoft, Software Engineer II - OneDrive and SharePoint

Jul 2021 – Dec 2024 | India

Technologies: C#, C++, .NET, MySQL, Kubernetes, Azure - Service Bus, Service Fabric, Data Factory, Entra ID

- Engineered Restricted Access Control for SharePoint, combining RBAC, ABAC, hierarchical ACL evaluation and AAD token claim checks to enforce security-group-scoped sharing policies across legacy C++ auth paths for 3.5M+ users.
- Built an Azure Data Factory-orchestrated Spark/Scala pipeline on Azure Data Lake to process 7M+ SharePoint policy events per run, generating tenant-level compliance reports and governance insights for admins.
- Developed an event-driven Outlook-to-Office Web comment sync service using Azure Service Bus, Service Fabric and S2S token authentication, enabling reliable @mention reply propagation across Microsoft 365 apps.
- Migrated Just-In-Time DLP to .NET 6 isolated Azure Functions v4 on AKS, using Docker, ACR, autoscaling, health probes and canary deployments to improve rollout safety and service scalability.

Amazon, Software Development Engineer Intern

Jan 2021 – Jun 2021 | India

Technologies: Java, SpringBoot, DynamoDB, AWS - Kinesis, SQS, SNS

- Built a Java DAG health-monitoring scheduler to parallelize checks across 100+ streaming pipelines; integrated CloudWatch alerting—cutting incident detection time from 30+ minutes to under 5 minutes.

TECHNICAL SKILLS

Languages & Databases: C, C++, C#, GoLang, Java, Python, Elasticsearch, MySQL, PostgreSQL, MongoDB

Frameworks & Libraries: .NET, Spring Boot, ReactJS, Node, Django, PyTorch, TensorFlow, Pandas, Kubernetes

Developer Tools: Git, CI/CD, VS Code, Visual Studio, PyCharm, IntelliJ, Docker, Claude Code

EDUCATION

University of Massachusetts, Amherst,

Jan 2025 – Dec 2026 | Massachusetts, USA

Master of Science in Computer Science, GPA: 4.0

Coursework: Reinforcement Learning, Machine Learning, Neural Networks, Systems for Data Science, Systems for Deep Learning, Algorithms for Data Science, Database Design & Implementation

Indian Institute of Information Technology, Sri City,

Jul 2017 – Jun 2021 | Chittoor, India

Bachelor's in Electronics and Communication Engineering

Women Techmakers Engineering Fellow 2019 - Selected among 100 women nationwide for a Google-supported bootcamp and scholarship.

PROJECTS

Watermarking LLM: Built steganographic LLM watermarking via top-p logits warping + McEliece ECC - batch encode/decode secret payloads in generated text across GPT-2 and Llama-family models.

RepOpt-3D: Trained a lightweight 2D-to-3D adapter to transfer frozen CLIP and DINOv2 features to 3D semantic segmentation on Matterport3D, reaching 48.89% mIoU vs. a 40.04% frozen baseline.